

Practical WS23: Planning Question

Time: 30 minutes

Name: **SOLUTIONS** ()

Class: 26S

Date:

[2023/P4/Q4]

Two of the sources of naturally-occurring background radiation are from buildings and from soil.

The amount of background radiation detected depends on the distance b from a building and the distance s from the soil.

The background count rate C is given by the equation

$$C = K b^p s^q$$

where K , p and q are constants.

Design an experiment to determine the values of p and q .

Assume that the investigation takes place outdoors close to a building and that you have a Geiger-Müller (GM) tube which counts individual particles.

Draw a diagram to show the arrangement of your apparatus. You should pay particular attention to

- (a) the equipment you would use,
- (b) the procedure to be followed,
- (c) the control of variables,
- (d) any precautions that should be taken to improve the accuracy of the experiment.

Note to Students:

This is a full planning question with a comprehensive set of solution.

You may need to modify your answers according to the requirements of questions with fewer marks.

As a guide, you may refer to the points that a question wants you to 'pay particular attention to'. This means that you do not need to include all the sections given in this set of solution, if the question does not require it. (For example: Not all planning questions require you to state precautions.)

For planning questions that follow a hands-on experiment, you may refer to the steps stated in the earlier parts instead of copying word for word into your answer.

However, these earlier parts do not include details such as the measuring instruments used, so you must state these explicitly in your answer. (For example, Q1(a) was an earlier part to time oscillations, and if you require this step in your planning, you can state in your answer: Carry out step Q1(a) using a stopwatch.)

Details that you should always include in your procedure, whether it is a full or short planning question, are:

- Independent variable and dependent variable (only 2 variables at any one time)
- Control variables
- Measuring instruments (to measure each variable)
- Equations need to calculate the variables to be plotted
- Steps to repeat to vary the independent variable to obtain at least 8 sets of readings (describe the method to vary the independent variable)
- Graph to be plotted to answer the objective of the question

Problem Definition

(You **must** identify the variables first, before writing your procedure. You need not write this as a separate section if you can describe the variables to be investigated clearly in your procedure.)

Experiment 1

Independent variable: Distance b from building

Dependent variable: Count rate C

Control variables:

- i. Distance s from soil
- ii. Background count rate due to other sources

Experiment 2

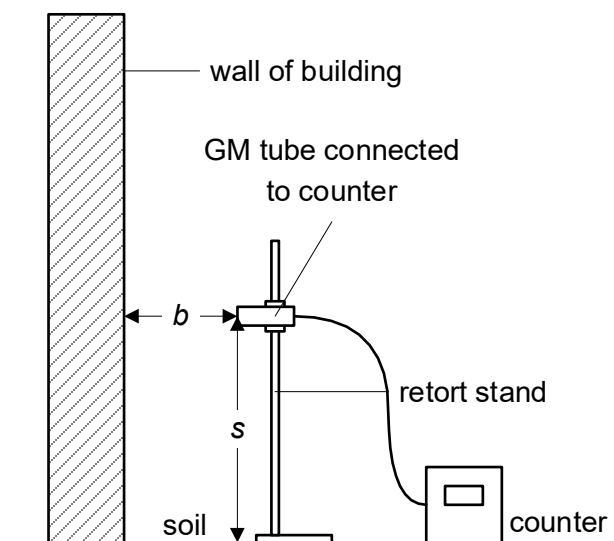
Independent variable: Distance s from soil

Dependent variable: Count rate C

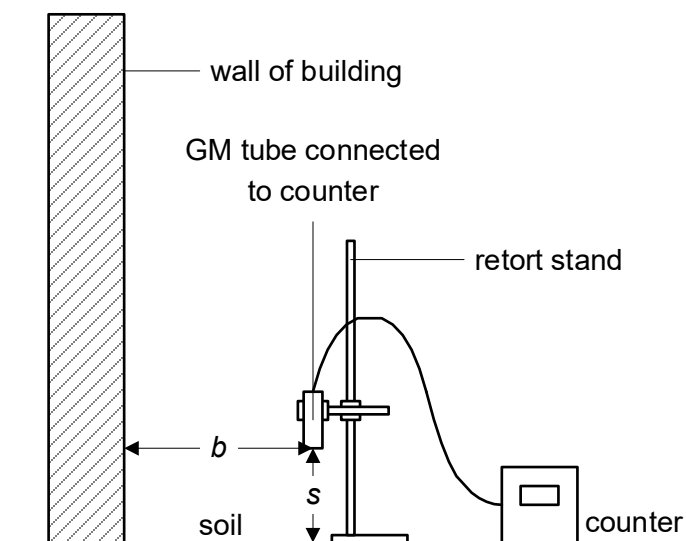
Control variables:

- i. Distance b from building
- ii. Background count rate due to other sources

Experimental Set-up



Variation with b
Fig. 1.1



Variation with s
Fig. 1.2

Procedure

1. Set up the apparatus as shown in Fig. 1.1.
2. The GM tube is directed towards the wall of a building.
3. Measure and record the distance b between the wall and the GM tube with a metre rule or measuring tape.

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4. Measure and record the distance s between the soil and the GM tube with a metre rule or measuring tape.
5. Connect the GM tube to the counter. Turn on the counter.
6. Measure and record the number of counts N detected in time t (e.g. 5 or 10 min) with a stopwatch.
7. The count rate is $C = \frac{N}{t}$.

Experiment 1

8. Repeat steps 3 to 7 by varying the distance b between the GM tube and the wall of the building for 8 sets of readings.
9. The distance s between the GM tube and the soil is kept constant by maintaining the height of the GM tube.

Experiment 2

10. Set up the apparatus as shown in Fig. 1.2. The GM tube is now directed towards the soil.
11. Repeat steps 3 to 7 by varying the distance s between the GM tube and the soil for 8 sets of readings.
12. The distance b between the GM tube and the wall of the building is kept constant by maintaining the position of the retort stand.

Analysis

$$C = K b^p s^q$$

Experiment 1

$$\lg C = p \lg b + \lg(Ks^q)$$

Plot a graph of $\lg C$ against $\lg b$.

If the relationship is true, a straight line will be obtained where p is the gradient and $\lg(Ks^q)$ is the vertical intercept.

Experiment 2

$$\lg C = q \lg s + \lg(Kb^p)$$

Plot a graph of $\lg C$ against $\lg s$.

If the relationship is true, a straight line will be obtained where q is the gradient and $\lg(Kb^p)$ is the vertical intercept.

Additional Details

1. Conduct preliminary experiments to determine a suitable duration such that a sufficiently large number of counts can be measured by GM tube. This is to minimise error due to random nature of radioactive decay.
2. Conduct preliminary experiment to determine a suitable range of distances from building or soil where the count rate varies sufficiently when varying either b or s .
3. Set $s \gg b$ when investigating the variation of C with b to ensure that the radiation detected is primarily due to the wall. Similarly, set $b \gg s$ when investigating the variation of C with s .
4. Measure the background count rate far away from any building and the soil to determine the radiation due to other sources. Subtract this background count rate from C .

[11]

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In this planning question, candidates were told that two sources of naturally occurring background radiation are from buildings and from soil. The amount of radiation detected depended on the distance b from a building and the distance s from the soil and that the background count rate C was given by $C = K b^p s^q$.

Candidates were asked to design an experiment to determine the values of p and q . Many candidates treated the soil and building as dangerous radioactive sources which required a protective suit. The soil was often shown as a separate source in a container. A large number of candidates stated that the background radiation should be measured a distance away from buildings and soil, then taken away from the count rate measured when directed at the building or the soil. A few candidates introduced another radioactive source and so described an incorrect experiment. Most candidates described the use of at least two pieces of equipment either shown on a diagram or mentioned in the text.

Better performing candidates showed good practice by listing the independent, dependent and control variables at the start of a given procedure. Weaker responses often did not mention the control variable or contained contradictions by treating either b or s as both the independent and control variable in a given procedure.

The full linearisation was rarely seen, with many candidates stopping short at $\lg C = p \lg b + \lg(Ks^q)$. Many candidates unnecessarily described the determination of K .

Suggested mark Scheme

Detailed Diagram	[max 1]
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GM tube connected to counter held by a retort stand with b and s shown clearly.	1
Procedures	[max 4]
Measure b and s with ruler or measuring tape.	1
Measure time t of data collection with stopwatch.	1
Measure count N with GM tube. Count rate: $C = \frac{N}{t}$	1
Correct orientation of GM tube in each experiment.	1
Repeat for a total of at least 10 sets of data by adjusting b or s .	1
Control of Variables	[max 2]
Experiment 1: Constant s by positioning GM tube a fixed distance from soil.	1
Experiment 2: Constant b by positioning GM tube a fixed distance from building.	1
Analysis	[max 2]
Experiment 1: Plot $\lg C$ against $\lg b$. Determine p from the gradient of the straight line.	1
Experiment 2: Plot $\lg C$ against $\lg s$. Determine q from the gradient of the straight line.	1
Additional Details	[max 2]
Conduct preliminary experiment to determine a suitable duration such that a sufficiently large number of counts can be measured by GM tube. This is to minimise error due to random nature of radioactive decay.	1
Conduct preliminary experiment to determine a suitable range of distances from building or soil where the count rate varies sufficiently when varying either b or s .	1
Set $s \gg b$ when investigating the variation of C with b to ensure that the radiation detected is primarily due to the wall. Similarly, set $b \gg s$ when investigating the variation of C with s .	1
Measure the background count rate far away from any building and the soil to determine the radiation due to other sources. Subtract this background count rate from C .	1